Anatomy of flowering Plants

Short notes and hidden lines

Anatomy of flowering Plants

Tissue – common origin and similar function

(not in syllabus -but should study)

Type of tissue – Meristematic- Tissue Parenchyma collenchyma, Sclerenchyma -Simple Permanent tissue and complex Permanent tissue – Xylem and Phloem

Meristematic Tissue- dense cytoplasm, Primary cell wall, Vacuole absent , Multiple plasmo desmatic connection, small size

Meristem	Nature	Position	Role
Apical	Primary	Root tip and shoot tip	Length increase, leaf formation
Intercalary	Primary	Node of monocot	Increase internode length and regenerate plant part once grazer eat tip
Meristem in Axillary position	Primary	Some meristem left behind from shoot apical	Form axillary bud which give branch or flower
Vascular cambium Cork cambium	Secondary	Lateral or cylindrical in dicot stem or Dicot root	Secondary xylem and Secondary phloem from vascular cambium Phellem and Phelloderm from cork cambium

Permanent tissue- Simple permanent tissue

Name of tissue	Parenchyma	collenchyma	Sclerenchyma
Shape	Oval,round,polygonal,spherical -Isodiametric and Elongated	Oval ,spherical and polygonal	Long, narrow cells and pits present (pit-no secondary wall or lignin)
Intercellular space	Present in some tissue- cortex and pith	Absent	Absent
Deposition	Cellulose	Cellulose, hemicellulose and Pectin	lignin
Location in plant	Epidermis,cortex,pith,endodermis,Meosphyll cell	Hypodermis of dicot stem and petiole of leaf	Xylem and phloem fibre,hypodermis of monocot stem and pericycle of dicot stem
Type	Mesophyll cell in leaf Spongy and palisade in Dicot leaf		Fibre- ligniifed,dead and lumen close **Sclerids** -highly thick wall, present in fruit wall of nuts,pear and sapota -pulp and legume seed coat ,tea leaf

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Complex Permanent Tissue –

Xylem – Cells – Xylem parenchyma ,Fibre ,Vessels, Tracheid (role-conduction of mineral in organic and inorganic form, hormones and water)

Features	Xylem parenchyma	Vessels	Sclereids	Fibre
Lignin	No	Yes	Yes	Yes -septate or aseptate
living	Yes	NO	NO	No
Pit	No	Yes	Yes	Yes
Role	Storage of oil, starch and tannins and radial conduction(ray parenchyma)	Conduction and mechanical	Conduction and mechanical	Mechanical

Feature	Vessels	tracheids
liginified	yes	yes
dead	yes	yes
Apoplast conduction	yes	yes
pitted	yes	yes
No of cells	Multicellular many vessel member	Unicellular
Perforation	Yes at end wall	No perforation
Lumen	Wider than tracheids	Narrower than vessels

Gymosperm -Vessels absent

Type of Xylem – Primary xylem -develop from Primary meristem and two type(Protoxylem and metaxylem) Secondary Xylem (Develop from vascular cambium ,present in dicot stem and dicot root)

Type of Xylem – Endarch -protoxylem towards pith- Dicot stem

Exarch- Protoxylem outside and metaxylem towards pith (Root both dicot and monocot)

Phloem

Type of cells – Phloem parecnhyma, phloem fibre,sieve tube and companion cell(role-Conduction of food and hormones and some time minerals

Gymnosperm -Sieve cell and Albuminous cell

Feature	Phloem parenchyma	Phloem fibre	Sieve tube	Companion cell
Living or dead	living	dead	living	living
lignified	no	Yes	No	No
Protoplast	yes	No	Yes	Yes

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Nucleus	Yes	No	No	Yes
Role	Storage Resins, mucilage and gums	Mechanical- Use as bast fibre in Flex,hemp and jute	End to end conduction of food	Maintain pressure gradient of sieve tube
Location	Absent in monocots	Secondary phloem	Both primary and secondary phloem	Primary and secondary phloem

Companion cell- nucleated, control activity of sieve tube

Sieve tube - Nucleus absent, Symplast, End wall have sieve pore or Perforation on sieve plate ,multiple connection with companion cell known as pit fields

Primary phloem-develop from primary meristem and two type -protophloem and metaphloem

Secondary phloem- develop from vascular cambium or secondary meristem

Feature	**Xylem**	**Phloem**
Living cells	1-parenchyma	3- except fibre
Lignified cells	3- except parecnhyma	1-Only fibre
Conduction of	Water,mineral,hormones	Food,mineral,hormone
Direction	Unidirectional	Bi or Multidirectional

Tissue system –

In any plant part Root, stem and leaf three type of tissues- Epidermal, Vascular and ground tissue

Epidermal Tissue – Parenchyma cells, elongated ,vacuole present

Structure on Epidermis	Root epidermis	Stem epidermis	Leaf epidermis
Root hair-unicellular	yes	no	no
Cuticle	No	Yes	Yes
Stomata	No	Yes(few)	Yes
Trichome	no	yes	Yes
Lenticel	Yes-Dicot	Yes- Dicot	No

Trichome – Multicellular, branched or unbranched , can be secretory , Prevent transpiration ,soft and stiff

Stomatal apparatus- **Guard cells + stomatal pore + subsidiary cell** (Guard cell -chloroplast ,Radially arrange microfibril, inner wall thick and elastic ,outer wall thin) Bean shape- Dicot, Dumbel shape- monocot

Vascular bundle- Cells of Xylem in group -Xylem bundle and Cells of Phloem in Group- Phloem bundle

Name of Bundle	**Feature**	**Present in**

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Radial	Xylem and phloem on different radius or alternate	Dicot root and monocot root
Conjoint	Xylem and Phloem on same line or radius	Leaf and stem
Conjoint close	No cambium present between Xylem and Phloem	Dicot leaf and Monocot stem
Conjoint Open	Intrafasicular cambium between Xylem and Phloem	Dicot stem

Ground tissue – Transverse Section – Vascular Bundle – Epidermal Tissue (All tissue except V.B and Epidermal tissue)

Transverse section (P=parenchyma, C= collenchyma, S = Sclerenchyma) - Root

Feature	Dicot Root	Monocot root
Epidermis (P)	Single layer, root hair present, Epiblema	same
Cortex (P)	Multilayer, intercellular space present ,innermost layer -Endodermis	same
Endodermis(P)	Barrel shape cell ,single layer, Suberin deposit on radial and tangential wall	same
Pericycle (P)	Thick wall -Give lateral root and cambium	No secondary growth
Pith (P)	Small , intercellular space present	Large
Vascular bundle	Radial, 2 to 6 , Exarch	Polyarch ,Radial ,Exarch
Conjunctive tissue	Present between Xylem and phloem	
Stele	All tissue internal to endodermis (pith, V.B, Pericycle, conjunctive tissue)	

Ground tissue – Cortex, pith, conjunctive tissue and pericycle(only parecnhyma)

Stem

Feature	Dicot stem	Monocot stem
Epidermis (P)	Single layer ,cuticle and stomata can present	
Cortex	Hypodermis-collenchyma + Cortical cell(p) + Endodermis(P) (starch sheath)	Cortex absent Hypodermis-Sclerenchyma
Pericycle	Semilunar patch ,sclerenchyma	Absent
Vascular bundle	Triangular, Conjoint, open , Equal in size , present in ring (Eustele)	Conjoint, close, unequal size, small outer and large inner, scattered, Bundle sheath cell-Sclerenchyma
Secondary Growth	Present	Absent
Pith	Present , intercellular space present	Absent

leaf

Feature	Dicot leaf	Monocot leaf
Name	Dorsiventral leaf	Isobilateral leaf
Upper surface	Adaxial	Abaxial
Lower surface	Abaxial	Abaxial(but not much difference on both surface
Stomata	Lower surface -more	Both surface equal

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Mesophyll cell	Close to upper-palisade and close to lower – Spongy	No distinction
Vascular bundle	Close,conjoint, large size in mid-vein ,size unequal	Close,conjoint,Except mid vein nearly equal size
Buliform cell	Absent	In grasses -on upper surface on vein ,large colorless cell ,when turgid- leaf surface exposed

Hidden lines –

1. **Monocot lack phloem parenchyma this means all section of monocots -roots ,stem and leaf not have phloem parenchyma but in ncert only for monocot stem it is mentioned**
2. **Monocot root also have epidermis ,root hair , cortex and endodermis**
3. **Tissue which is part of ground tissue and stele both in dicot root is – Pericycle , pith and conjunctive tissue**
4. **Cuticle present in stem and leaf both**
5. **All epidermal cells including subsidiary cells lack chloroplast ,only guard cells have chloroplast**
6. **Hypodermis of dicot stem have collenchyma , give strength and also have chloroplast in young green stem**
7. **Water filled cavity in monocot stem present in xylem in place of protoxylem where xylem parenchyma absent**
8. **Bulliform cells are colourless also change shape like guard cell but lack chloroplast**
9. **In monocot root it is mentioned vascular bundle more than six for this they are polyarch and dicot root 2-4 xylem and phloem**
10. **Sclerenchyma mostly part of xylem and phloem as fibre but also present in dicot stem -pericycle and monocot stem – hypodermis and bundle sheath cell**
11. **Phloem fibre present only in secondary phloem this means only section of dicot stem,dicot root have phloem fibre**
12. **Epidermal tissue have parenchyma cells with elongated shape with no space but cortex mostly have cells with space**
13. **Stem have cortex with collenchyma hypodermis -no space in cells , cortical cells - space present , endodermis -have starch grain come from phloem**
14. **Dicot stem vascular bundle are conjoint,open, endarch and equal size and in ring but in monocot stem vascular bundle is conjoint close, unequal size and scatterd**
15. **Unequal size vascular bundle present in monocot stem and dicot leaf**
16. **Since dicot stem have reticulate venation ,where veins size are unequal so vascular bundle unequal size**

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